

the REST API:

```
import httpx

async with httpx.AsyncClient() as client:
    result = await client.post(
        "http://localhost:8080/api/v1/discovery/analyze",
        json={
            "traffic_data": ["<base64_packet_1>", "<base64_packet_2>"],
            "options": {
                "confidence_threshold": 0.7,
                "enable_adaptive_learning": True
            }
        }
    )

    protocols = result.json()["discovered_protocols"]
    for p in protocols:
        print(f"{p['name']}: {p['confidence']:.0%} confidence")
```

The nine built-in compliance frameworks

QBITEL Bridge includes automated compliance engines for nine regulatory frameworks: SOC 2, GDPR, HIPAA, PCI-DSS 4.0, ISO 27001, NIST 800-53, BASEL-III, NERC-CIP, and FDA 21 CFR Part 11. Audit-ready reports can be generated in under 10 minutes. The compliance agents continuously monitor for policy violations and produce evidence trails suitable for external audits.

Observability from Day One

QBITEL Bridge exports Prometheus metrics and OpenTelemetry traces out of the box. Every agent task, protocol discovery request, crypto operation, and cache hit is measured:

```
protocol_discovery_requests_total{status="success"} 1547
protocol_discovery_duration_seconds_bucket{le="0.15"} 892
qbitel_agent_tasks_total{agent_type="incident_response"} 423
pqc_operations_total{algorithm="ml-kem-1024", operation="encapsulate"} 98412
```

Grafana dashboard configurations are included in *ops/grafana-dashboards/*. Jaeger provides distributed tracing across the Python, Rust, and Go components.

Performance at a glance

- Protocol discovery accuracy: 89%+ on first pass
- Discovery time: 2 to 4 hours on a typical enterprise segment
- Parser generation throughput: 50,000+ messages per second
- Kafka event streaming: 100,000+ messages per second
- PQC encryption overhead: Under 1 millisecond
- Security decision time: Under 1 second
- Autonomous threat response rate: 78%
- xDS proxy support: 1,000+ concurrent proxies
- eBPF container monitoring: 10,000+ containers
- API gateway P99 latency: Under 25 milliseconds

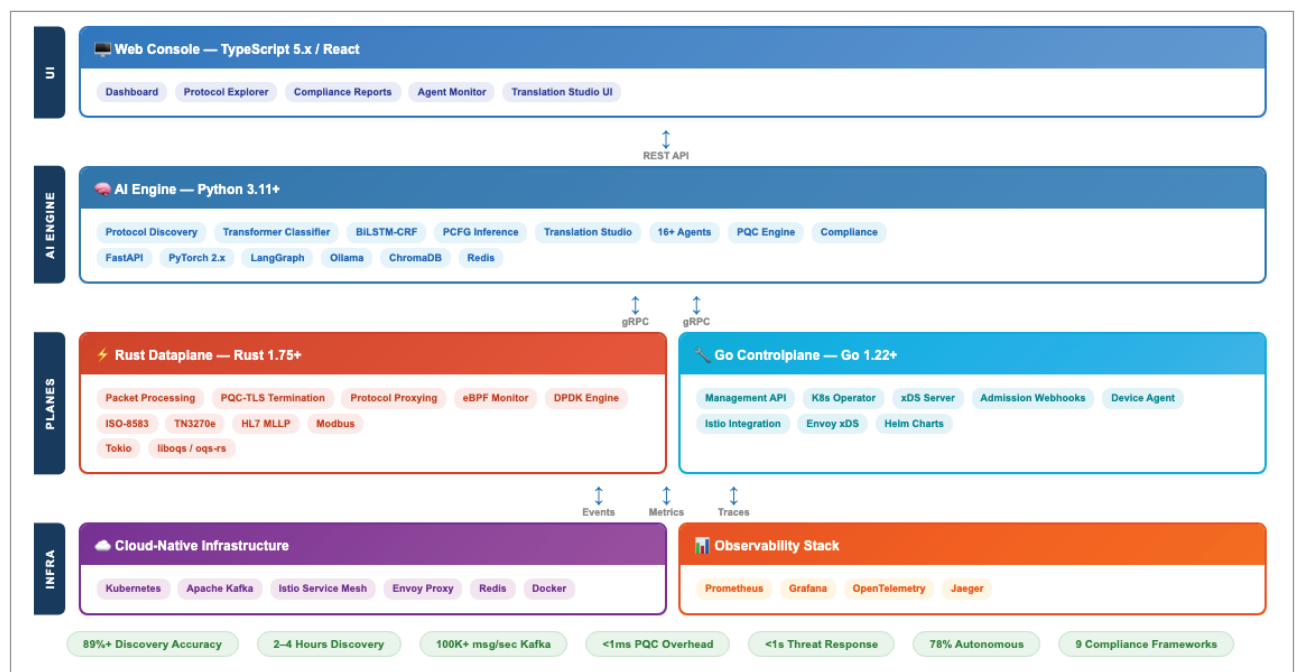


Figure 3: Four-layer system architecture